

Explainable Credit Risk Prediction Using Machine Learning

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Abstract:

Credit risk assessment is a critical task in the financial sector, where incorrect decisions can lead to significant economic losses.

Traditional machine learning models often act as black boxes, making it difficult to justify predictions. This project proposes an explainable credit risk prediction framework using Logistic Regression, Random Forest, and XGBoost models. To ensure transparency, Explainable Artificial Intelligence (XAI) techniques using SHAP are applied for both global and local interpretation. Experimental results demonstrate that ensemble models outperform baseline approaches, while SHAP explanations enhance trust and accountability in decision-making.

Chapter_1:Introduction

Credit risk prediction aims to determine whether a loan applicant is likely to default. With the increasing availability of data, machine learning techniques have become widely adopted for this task. However, many high-performing models lack interpretability, which is essential in regulated domains such as finance.

This project focuses on building a complete machine learning pipeline for credit risk prediction while emphasizing explainability. By integrating SHAP-based explanations, the model provides insights into both overall feature importance and individual predictions, enabling transparent and responsible AI-driven decision-making.

Chapter 2: Related Work

Previous studies have explored the use of Logistic Regression, Decision Trees, Random Forests, and Gradient Boosting models for credit risk assessment. Logistic Regression is commonly used due to its simplicity and interpretability, while ensemble models such as Random Forest and XGBoost have shown superior predictive performance.

Recent research highlights the importance of Explainable AI (XAI) techniques to interpret complex models. SHAP has emerged as a unified framework for explaining predictions by quantifying the contribution of each feature. This project builds upon these approaches by combining strong predictive models with robust explainability.

Chapter 3: Dataset Description

The dataset used in this project is a structured credit risk dataset containing 200 instances. Each instance represents a loan applicant with the following attributes:

- Age
- Income
- Loan Amount
- Employment Years
- Credit History
- Default (Target Variable)

The target variable is binary, where 0 indicates no default and 1 indicates default. The dataset exhibits class imbalance, reflecting real-world credit risk scenarios.

Chapter 4: Methodology

The methodology follows a structured pipeline:

4.1 Data Preparation

- Data loading and inspection
- Missing value handling
- Feature scaling using StandardScaler
- Train–test split with stratification

4.2 Model Development

- Logistic Regression as a baseline model
- Random Forest for non-linear learning
- XGBoost for boosted ensemble learning

4.3 Model Evaluation

- Accuracy
- Precision, Recall, and F1-score
- Confusion matrix analysis

4.4 Explainability

- SHAP summary plots for global interpretation
- SHAP waterfall plots for local explanation

Chapter 5: Results and Evaluation

The results indicate that ensemble models outperform Logistic Regression in terms of predictive performance. XGBoost achieved the highest accuracy and F1-score, followed closely by Random Forest. Logistic Regression served as a strong baseline model with good interpretability.

Confusion matrix analysis revealed the importance of minimizing false negatives, as misclassifying a defaulting applicant can lead to financial loss.

Chapter 6: Explainability Analysis

SHAP-based explainability revealed that loan amount is the most influential factor increasing default risk. In contrast, income and credit history significantly reduce the probability of default. Employment years also contribute to stabilizing predictions.

Local explanations demonstrated how individual features contribute positively or negatively to specific predictions, ensuring transparency at the decision level.

Chapter 7: Conclusion

This project successfully demonstrates an explainable credit risk prediction framework that balances predictive performance and interpretability. By integrating SHAP with machine learning models, the system provides transparent and trustworthy predictions. The framework is suitable for real-world financial applications where accountability and fairness are essential.

Chapter 8: Future Work

Future enhancements may include:

- Incorporation of real-world banking datasets
- Advanced handling of categorical features
- Fairness and bias analysis
- Deployment as a web-based decision support system

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